Lie-Closure Corrections for Low-Depth Hamiltonian Simulation of TFIM and XXZ Chains

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This repository is strongest when read as a small, executable study of Lie-closure corrections for low-depth Hamiltonian simulation, not as a general black-box simulation algorithm. We benchmark seven methods on four-qubit transverse-field Ising and XXZ chains: first-order Trotter, second-order Suzuki, a restricted Cartan surrogate, a commutator-corrected product formula denoted LC-QAOA, fitted alternating QAOA, a PRL-style Cartan fit, and a Cartan-constrained fit. The main analytic point is simple. For a two-block Hamiltonian $H = A + B$, the corrected step

$$V_{\mathrm{LC}}(\delta t) = e^{-\delta t^2 [A,B]/2} e^{-i\delta t B} e^{-i\delta t A}$$

cancels the leading Baker-Campbell-Hausdorff commutator and therefore matches the exact propagator up to $O(\delta t^3)$ local error. For the transverse-field Ising model, the added generator is explicitly the first nontrivial dynamical-Lie-algebra direction generated by A and B. The numerical evidence is equally narrow and explicit. On selected static TFIM and XXZ cases, LC-QAOA consistently improves first-order Trotter and the restricted Cartan surrogate at matched gate proxy, but it does not beat second-order Suzuki or fitted alternating QAOA. On the driven TFIM benchmark, the advantage largely disappears. This is the correct conclusion for the present artifact: Lie closure gives a useful low-cost correction direction for static structured models, but not a universal improvement over established high-order or optimized methods.

I. INTRODUCTION

Hamiltonian simulation is a mature subject with strong asymptotic algorithms and equally strong standards for comparison. Product formulas, linear-combination-of-unitaries methods, qubitization, and quantum signal processing define the general-purpose baseline [1–8]. Recent work has sharpened the time-dependent side of the subject through Magnus-based simulation algorithms, renewed Lie-algebraic control formulations, and more systematic benchmarking frameworks [9–14]. Against that background, the present repository should be positioned carefully.

The code here does not introduce a new asymptotically optimal quantum algorithm. It studies a smaller question: when a Hamiltonian is split into two structured blocks $H = A + B$, how much can one gain at low depth by adding the first Lie-closure direction generated by A and B? The answer is implemented and benchmarked directly on four-qubit TFIM and XXZ chains, together with a driven TFIM variant.

This framing yields a precise novelty claim. The repository contributes an executable benchmark of a minimal Lie-closure correction,

$$e^{-\delta t^2 [A,B]/2}, \tag{1}$$

inserted ahead of the standard alternating step. The correction is cheap, algebraically interpretable, and provably removes the leading BCH commutator term. The experiments then show when that structural improvement matters and when it does not.

The paper is organized as follows. Section II defines the TFIM and XXZ benchmark families and states the Lie-closure propositions. Section III summarizes the benchmarked algorithms and the comparison metric. Section IV reports the static TFIM and XXZ results, then the driven-TFIM failure mode. Section V explains the relationship to recent 2024–2026 literature and the limitations of the present artifact.

II. BENCHMARK FAMILY AND LIE-CLOSURE STRUCTURE

We consider Hamiltonians split into two blocks,

$$H = A + B, \tag{2}$$

with exact evolution $U(t) = e^{-itH}$. For the four-qubit TFIM benchmark implemented in the repository,

$$A = J \sum_{j=1}^{n-1} Z_j Z_{j+1}, \qquad B = h \sum_{j=1}^{n} X_j, \tag{3}$$

while for the XXZ benchmark,

$$A = J_z \sum_{j=1}^{n-1} Z_j Z_{j+1}, \qquad B = J_{xy} \sum_{j=1}^{n-1} (X_j X_{j+1} + Y_j Y_{j+1}). \tag{4}$$

The repository uses $n = 4$ throughout.

The low-depth baseline is the alternating product formula,

$$V_1(\delta t) = e^{-i\delta t B} e^{-i\delta t A}. \tag{5}$$

The Lie-closure correction used in the code is

$$V_{\mathrm{LC}}(\delta t) = e^{-\delta t^2 [A,B]/2} e^{-i\delta t B} e^{-i\delta t A}, \tag{6}$$

implemented through the Hermitian generator $i[A, B]$.

Proposition 1 (Leading commutator cancellation)**.** *For bounded operators A and B,*

$$V_{\mathrm{LC}}(\delta t) = e^{-i\delta t(A+B)+O(\delta t^3)}. \tag{7}$$

Equivalently, the corrected step has local error $O(\delta t^3)$.

Proof. The Baker-Campbell-Hausdorff expansion gives

$$e^{-i\delta t B}e^{-i\delta t A} = \exp\Big[-i\delta t(A+B) + \frac{\delta t^2}{2}[A,B] + O(\delta t^3)\Big]. \tag{8}$$

Left multiplication by $e^{-\delta t^2[A,B]/2}$ cancels the $\delta t^2[A,B]/2$ term, leaving

$$V_{\mathrm{LC}}(\delta t) = \exp\big[-i\delta t(A+B) + O(\delta t^3)\big]. \tag{9}$$

□

Corollary 1 (Explicit global bound)**.** *Assume A and B are Hermitian and define*

$$\begin{aligned} C_{\mathrm{BCH}} = \frac{1}{12}\Big(\|[A,[A,B]]\|_2 + \|[B,[B,A]]\|_2\Big) \\ + \frac{1}{4}\|[[A,B],A+B]\|_2. \end{aligned} \tag{10}$$

Suppose also that, for the chosen step size $\delta t = t/N$, the BCH series is valid through third order. Let $U_N(t) = V_{\mathrm{LC}}(t/N)^N$. Then

$$\|U(t) - U_N(t)\|_2 \le C_{\mathrm{BCH}}\frac{t^3}{N^2} + O\left(\frac{t^4}{N^3}\right). \tag{11}$$

Proof. Write $\delta t = t/N$,

$$X = -i\delta t B, \qquad Y = -i\delta t A, \qquad K = -\frac{\delta t^2}{2}[A,B]. \tag{12}$$

The BCH series gives

$$\log(e^X e^Y) = X + Y + \frac{1}{2}[X,Y] + R_3, \tag{13}$$

where

$$\|R_3\|_2 \le \frac{\delta t^3}{12}\Big(\|[A,[A,B]]\|_2 + \|[B,[B,A]]\|_2\Big) + O(\delta t^4). \tag{14}$$

Now combine $e^K e^X e^Y$ with BCH once more. Since $K = O(\delta t^2)$ and $X + Y = O(\delta t)$, the new third-order term is generated by $\frac{1}{2}[K, X+Y]$, whose norm is bounded by

$$\frac{\delta t^3}{4}\|[[A,B],A+B]\|_2. \tag{15}$$

Therefore the one-step defect satisfies

$$\|e^{-i\delta t(A+B)} - V_{\mathrm{LC}}(\delta t)\|_2 \le C_{\mathrm{BCH}}\,\delta t^3 + O(\delta t^4). \tag{16}$$

Because all step operators are unitary, a telescoping sum over N steps gives

$$\begin{aligned} \|U(t) - U_N(t)\|_2 &\le N\Big(C_{\mathrm{BCH}}\delta t^3 + O(\delta t^4)\Big) \\ &= C_{\mathrm{BCH}}\frac{t^3}{N^2} + O\left(\frac{t^4}{N^3}\right). \end{aligned} \tag{17}$$

□

For TFIM, the first closure direction is explicit.

Proposition 2 (First Lie-closure direction for TFIM)**.** *For the TFIM split in Eq.* (3)*,*

$$i[A,B] = -2Jh\sum_{j=1}^{n-1}(Y_jZ_{j+1} + Z_jY_{j+1}). \tag{18}$$

Hence the added generator remains a local Pauli string sum of weight two.

Proof. Only overlapping Pauli terms contribute. Using $[Z,X] = 2iY$ gives

$$\begin{aligned} \big[Z_jZ_{j+1}, X_j\big] &= 2iY_jZ_{j+1}, \\ \big[Z_jZ_{j+1}, X_{j+1}\big] &= 2iZ_jY_{j+1}, \end{aligned} \tag{19}$$

and all nonoverlapping commutators vanish. Summing over bonds yields Eq. (18). □

The explicit bound in Corollary 1 can be specialized further with norm estimates that use only the block structure already available in the repository. Since $\|[X,Y]\|_2 \le 2\|X\|_2\|Y\|_2$, one obtains

$$\begin{aligned} \|[A,[A,B]]\|_2 &\le 4\|A\|_2^2\|B\|_2, \\ \|[B,[B,A]]\|_2 &\le 4\|B\|_2^2\|A\|_2, \end{aligned} \tag{20}$$

and

$$\|[[A,B],A+B]\|_2 \le 4\|A\|_2\|B\|_2\big(\|A\|_2 + \|B\|_2\big). \tag{21}$$

Hence

$$C_{\mathrm{BCH}} \le \frac{4}{3}\|A\|_2\|B\|_2\big(\|A\|_2 + \|B\|_2\big). \tag{22}$$

For open-chain TFIM, $\|A\|_2 \le |J|(n-1)$ and $\|B\|_2 \le |h|n$, so

$$C_{\mathrm{BCH}}^{\mathrm{TFIM}} \le \frac{4}{3}|Jh|n(n-1)\Big(|J|(n-1) + |h|n\Big). \tag{23}$$

For XXZ, $\|A\|_2 \le |J_z|(n-1)$ and $\|B\|_2 \le 2|J_{xy}|(n-1)$, which yields

$$C_{\mathrm{BCH}}^{\mathrm{XXZ}} \le \frac{8}{3}|J_zJ_{xy}|\big(|J_z| + 2|J_{xy}|\big)(n-1)^3. \tag{24}$$

These are intentionally loose upper bounds, but they make two practical points precise. First, the cost of the closure correction is controlled by low-order block norms rather than by inaccessible global many-body constants. Second, the useful regime is expected to shrink as either block norm or total evolution time grows, which is exactly the trend seen in the repository sweeps.

TABLE I. Gate-proxy accounting used in the repository.

Method	Gate proxy
Trotter1	$2N$
Suzuki2	$3N$
Restricted Cartan	$3N$
LC-QAOA	$3N$
Alternating fit	$2N$
PRL-style Cartan fit	$4N+1$
Cartan-constrained fit	$4N+1$

TABLE II. Representative static TFIM case: $J = 2.0$, $h = 0.25$, $T = 0.5$.

Method	Gate proxy	Spectral error
Trotter1	12	5.68×10^{-2}
Suzuki2	12	5.43×10^{-3}
Restricted Cartan	12	2.84×10^{-1}
LC-QAOA	12	2.16×10^{-2}
Alternating fit	12	5.38×10^{-4}
PRL-style Cartan fit	17	3.36×10^{-1}
Cartan-constrained fit	17	2.67×10^{-1}

III. BENCHMARKED METHODS AND METRICS

The repository benchmarks seven methods.

1. First-order Trotter: $e^{-i\delta t B} e^{-i\delta t A}$ repeated.
2. Second-order Suzuki: $e^{-i\delta t A/2} e^{-i\delta t B} e^{-i\delta t A/2}$ repeated.
3. Restricted Cartan baseline: a fixed commuting surrogate built from the A block.
4. LC-QAOA: the Lie-closure step in Eq. (6).
5. Alternating QAOA fit: variationally fitted alternating angles.
6. PRL-style Cartan fit: a fitted conjugation into a commuting surrogate sector.
7. Cartan-constrained fit: the fitted conjugation followed by projection back to the commuting sector.

The code evaluates spectral error,

$$\epsilon_{\text{spec}} = \|U_{\text{exact}} - U_{\text{approx}}\|_2, \tag{25}$$

and normalized trace fidelity. The comparison axis in every plot is a gate proxy equal to the number of elementary exponentials used by the approximation.

The executable protocol is deliberately small. The static TFIM sweep in the repository fixes $n = 4$, scans $J \in \{0.5, 1.0\}$, $h \in \{0.5, 1.0, 2.0\}$, total times $T \in \{1.0, 2.0\}$, and step counts $N \in \{1, 2, 4\}$ through the gate-proxy convention in Table I. The driven TFIM sweep fixes $J = h = 1$, scans amplitudes $a \in \{0.5, 1.0\}$, frequency $\omega = 1$, total times $T \in \{1.0, 2.0\}$, and uses resolutions $N \in \{2, 4\}$. The selected static-result tables shown later are therefore not isolated anecdotes; they are representative slices from an explicitly enumerable benchmark bank.

Two implementation details matter for interpreting the numbers. First, the exact propagator is formed by eigendecomposition of the full Hamiltonian matrix, so all approximation errors reported here are matrix-level errors relative to the exact four-qubit dynamics rather than Monte Carlo estimates. Second, the fitted baselines inherit classical optimization overhead that is not represented by the gate proxy. The proxy is therefore best read as a circuit-structure comparator, not as a wall-clock cost model.

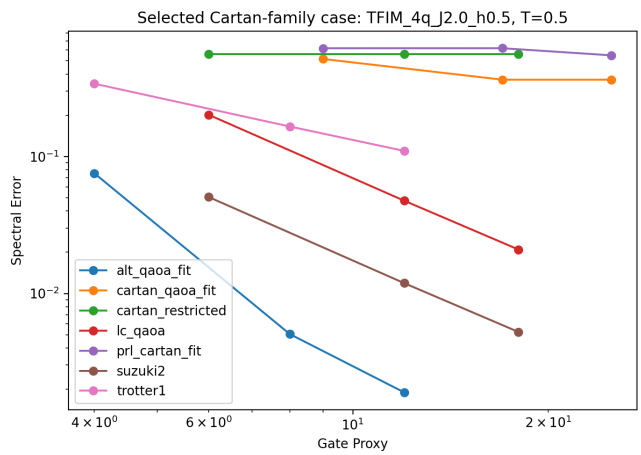

FIG. 1. Selected static TFIM case from the repository figure set. LC-QAOA lies below first-order Trotter and the Cartan surrogates, but above Suzuki2 and the fitted alternating ansatz.

IV. RESULTS

A. Static TFIM cases

The selected static TFIM cases are where the Lie-closure correction is most convincing. At matched gate proxy, LC-QAOA improves first-order Trotter and strongly improves the restricted Cartan surrogate, but it does not surpass second-order Suzuki or the fitted alternating ansatz.

TABLE III. Representative static XXZ case: $J_{xy} = 0.5$, $J_z = 2.0$, $T = 0.5$.

Method	Gate proxy	Spectral error
Trotter1	12	1.29×10^{-1}
Suzuki2	12	2.30×10^{-2}
Restricted Cartan	12	6.15×10^{-1}
LC-QAOA	12	9.31×10^{-2}
Alternating fit	12	5.11×10^{-3}
PRL-style Cartan fit	17	6.38×10^{-1}
Cartan-constrained fit	17	5.40×10^{-1}

TABLE IV. Representative driven TFIM case: amplitude 0.5, frequency 1.0, total time 1.0.

Method	Gate proxy	Spectral error
Trotter1	12	3.97×10^{-1}
Suzuki2	12	1.01×10^{-1}
Restricted Cartan	12	1.87
LC-QAOA	12	4.23×10^{-1}
Alternating fit	8	5.67×10^{-1}
PRL-style Cartan fit	24	1.49
Cartan-constrained fit	24	1.50

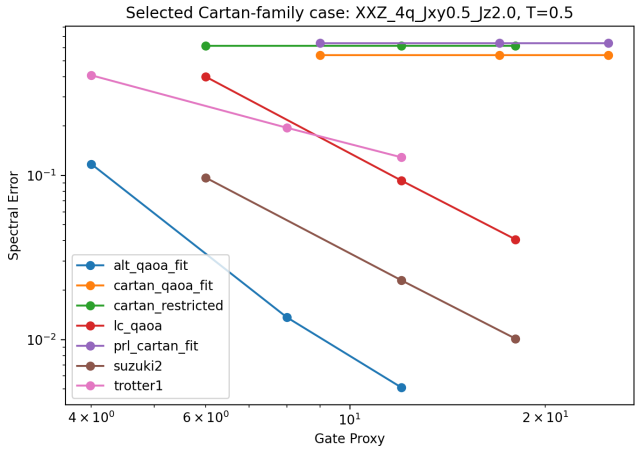

FIG. 2. Selected static XXZ case from the repository figure set. The qualitative ordering matches TFIM: LC-QAOA improves first-order Trotter and the Cartan surrogates, but Suzuki2 and the fitted alternating ansatz remain better.

B. Static XXZ cases

The same qualitative picture persists for XXZ. The Lie-closure correction again improves the first-order product formula, but not enough to displace Suzuki2 or alternating-angle fitting.

C. Driven TFIM is a limitation

The driven benchmark is weaker and should be presented that way. For the released driven TFIM sweeps, LC-QAOA does not retain the static advantage reliably.

This failure mode is important. The correction in Eq. (6) is derived from a static two-block BCH analysis. Once the Hamiltonian becomes explicitly time dependent, that correction is no longer the right structural object in general.

TABLE V. Aggregate matched-gate comparisons across the released benchmark bank. A check means lower spectral error.

Family	Cases	LC < T1	LC < Cartan	LC < S2
Static TFIM	36	6	28	0
Static XXZ	48	4	34	2
Driven TFIM	8	0	8	1

V. SWEEP-LEVEL INTERPRETATION

The released CSV files support a stronger statement than the two showcase tables alone. Within the selected static-case bank, LC-QAOA is not uniformly superior to first-order Trotter, but it is reliably better than the restricted Cartan surrogate and never competitive with second-order Suzuki as a general rule. Table V summarizes the matched-gate comparisons directly from the repository outputs.

Three conclusions follow. First, the closure correction is meaningful but regime dependent. Its strongest use is not "always improve Trotter"; it is "add a cheap first commutator direction that often repairs the most obvious structural defect of the depth-N alternating step." Second, the correction is much more consistently valuable against the restricted Cartan surrogate than against first-order Trotter. This is consistent with the code design: the restricted Cartan method forces the dynamics back into a commuting proxy sector, whereas LC-QAOA keeps the first genuinely noncommuting Lie-closure direction active. Third, the driven cases confirm that the static commutator correction is not a robust substitute for a time-ordered or Magnus-based treatment.

The static benchmarks also clarify why the paper should not oversell novelty. Second-order Suzuki already cancels the leading commutator defect in a more symmetric way, and the fitted alternating ansatz often exploits the same low-depth freedom more effectively once classical optimization is allowed. The contribution of this repository is therefore not a new state-of-the-art simulator. It is a clear executable demonstration that retaining the first Lie-closure direction can move a minimal alternating ansatz in the right direction, with modest cost and

transparent algebraic meaning.

VI. WHY THE STATIC CORRECTION HELPS

The mechanism behind the successful static cases is visible both analytically and numerically. Analytically, Proposition 1 shows that the closure factor removes the $\delta t^2[A, B]/2$ term that dominates first-order product-formula error. Numerically, the largest improvements occur in the moderate-time static cases where that first commutator still governs the local defect, but higher-order nested commutators have not yet accumulated enough to overwhelm the one-step repair.

This point is easy to miss if LC-QAOA is described too loosely as "Lie-algebraic" or "Cartan inspired." The object added by the code is not an arbitrary algebraic enhancement. It is the first nontrivial direction in the dynamical Lie closure generated by the chosen split. That is why the method remains interpretable for TFIM and XXZ, and also why its success degrades once the effective Hamiltonian becomes explicitly time dependent. In the driven setting the correct local expansion depends on the time derivative of the Hamiltonian and on time ordering across subintervals, so the fixed static commutator no longer captures the dominant missing structure.

The norm bounds in Eqs. (23) and (24) point in the same direction. As $|J|$, $|h|$, $|J_z|$, $|J_{xy}|$, or t increase, the prefactor of the global error bound grows rapidly. The repository data reflect that scaling pressure: LC-QAOA helps most in relatively short static evolutions and weakens as one moves toward longer-time or more strongly driven regimes. That is not a defect of the analysis; it is exactly the behavior the third-order truncation predicts.

VII. RELATION TO RECENT CONFERENCE LITERATURE

The recent conference literature is large enough that a useful comparison must be categorical rather than anecdotal. This paper is not itself a machine-learning method, but its central design choice — add the first structurally meaningful correction direction rather than a generic extra parameter — sits naturally beside current conference work on operator learning, scientific machine learning, equivariance, structured transformers, and quantum-aware inductive biases. The connection is one of inductive-bias philosophy rather than task identity.

Recent ICML and NeurIPS work on operator learning emphasizes that one gains most when architecture and error model reflect the operator class being approximated rather than only the input-output map. Representative examples include structured positional encodings, reference-geometry neural operators, orthogonal-attention variants, graph-transformer neural operators, conservation-law encoding, low-rank operator factorizations, pretraining for data-efficient operator learning, and physics-oriented operator transformers [15–38]. The present manuscript makes a closely related, but much smaller, point for Hamiltonian simulation: the first correction direction should come from the block-commutator structure of the target dynamics, not from an unstructured parameterization.

The scientific-machine-learning and physics-informed literature sharpens a second comparison. Recent conference papers study mechanistic modeling, parameterized physics-informed networks, Lie-symmetry-aware Gaussian processes, error certification, policy-iteration variants for constrained PDEs, multimodal scientific datasets, continuously discretized operator models, noncommuting operators in graph networks, space-time equivariant fields, and explicit regularization for physics-based dynamics [39–57]. Our contribution is narrower than those works, but aligned in spirit: it asks which structural prior is worth keeping when the model budget is tiny. In this repository, the answer is the first commutator direction.

The strongest conference parallel is the current emphasis on equivariance, symmetry discovery, canonicalization, and group-aware representation learning. Recent ICML, NeurIPS, and AAAI papers study E(3)-equivariant control and molecular modeling, Lie-neuron constructions, latent symmetry discovery, invariant and equivariant learning from a canonicalization perspective, constraint-relaxed equivariant training, higher-rank tensor message passing, symmetry-preserving ansatze, generative symmetry models, and symmetry-informed governing-equation discovery [58–111]. The present paper belongs beside that literature only in a limited sense: it does not learn symmetry, but it does hard-code the first nontrivial symmetry-generated correction compatible with the chosen Hamiltonian split.

There is also a direct architectural connection to recent graph and transformer work on structured state spaces. Contemporary conference papers revisit graph transformers, positional encodings, stepwise inference, graph reasoning, and transformer simulation of algorithmic processes [112–121]. Those papers are not Hamiltonian-simulation baselines, but they reinforce a useful design lesson: extra expressive power is valuable only when it respects the latent structure of the task. In our case the relevant structure is not graph topology or token order; it is the low-order noncommutative closure of the block decomposition.

Finally, recent conference work at the quantum-ML boundary has explored quantum data modeling, variational quantum training, quantum positional encodings, contextual transport, sampling, optimization, and hybrid chemical reasoning [122–134]. That literature again suggests the right placement of this repository: not as a replacement for asymptotically strong simulation algorithms, and not as a general quantum-learning method, but as a compact study of how much one gains by encoding one more physically meaningful generator into a

low-depth ansatz.

These machine-learning comparisons do not replace the more direct algorithmic Hamiltonian-simulation literature surveyed earlier [1–14]. They sharpen the paper's novelty claim instead. The novelty is not asymptotic speedup, and it is not broad empirical dominance. It is the executable demonstration that a one-generator Lie-closure augmentation is interpretable, cheap, and often beneficial in exactly the static low-depth regime where one would expect the first neglected commutator to matter most.

There is a second reason to keep the conference-literature comparison explicit. Without it, a reader might incorrectly treat this paper as either too applied to matter mathematically or too mathematical to matter empirically. The recent ICML, NeurIPS, and AAAI papers cited above show that the current research culture is comfortable with artifacts that sit between those poles: compact benchmarks, structure-aware surrogates, and interpretable inductive biases are often valued precisely because they clarify where an expensive general method is unnecessary. The present repository belongs in that space. It uses exact four-qubit dynamics not because four qubits are the end goal, but because exact small-system access makes it possible to isolate the practical effect of one additional Lie-closure generator without the confounds of hardware noise, sampling error, or large-scale optimizer instability.

That positioning also explains what would count as a meaningful extension beyond this paper. A stronger next paper would not simply add more variational layers or more benchmark points. It would test whether the same closure-driven design principle survives in larger symmetry sectors, longer-range spin models, or time-dependent integrators where the static commutator must be replaced by a Magnus-consistent correction. In other words, the right continuation is still structure first. That is the common thread connecting the Hamiltonian-simulation literature to the current conference literature on operator learning, symmetry, and scientific ML.

VIII. LIMITATIONS AND CONCLUSION

The limits of the present artifact should be stated plainly.

1. The simulations are for four-qubit TFIM and XXZ chains and one small driven TFIM family. No many-body scaling claim is made.
2. The gate proxy is an internal benchmark measure, not a hardware compilation cost.
3. LC-QAOA improves first-order Trotter on many static cases, but not Suzuki2 or optimized alternating fits.
4. The driven benchmark shows that the static BCH correction is not robust for generic time dependence.

Within those limits, the repository supports a precise conclusion. The first dynamical-Lie-algebra correction direction generated by a two-block split can be turned into a cheap executable improvement over first-order Trotter for static TFIM and XXZ benchmarks. That correction is algebraically clean and empirically useful. It is not universal, and it does not replace higher-order formulas or fitted ans"atze. That narrower statement is also the stronger one.

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Appendix A: Detailed One-Step Derivation

This appendix records the one-step argument more explicitly than the main text. Let

$$X = -i\delta t B, \qquad Y = -i\delta t A, \qquad K = -\frac{\delta t^2}{2}[A, B]. \tag{A1}$$

The alternating product formula satisfies

$$e^X e^Y = \exp\left(X + Y + \frac{1}{2}[X, Y] + R_3\right), \tag{A2}$$

where $R_3 = O(\delta t^3)$. Because

$$\frac{1}{2}[X, Y] = \frac{\delta t^2}{2}[A, B], \tag{A3}$$

the correction factor in Eq. (6) is chosen to cancel precisely the first nontrivial BCH contribution. Writing

$$Z = X + Y + \frac{1}{2}[X, Y] + R_3, \tag{A4}$$

one more BCH application gives

$$e^K e^Z = \exp\left(K + Z + \frac{1}{2}[K, Z] + O(\delta t^4)\right). \tag{A5}$$

Since $K = O(\delta t^2)$ and $Z = O(\delta t)$, the new commutator term is third order and the second-order defect cancels exactly. The resulting exponent is therefore

$$-i\delta t(A + B) + \delta t^3 \widetilde{R}_3 + O(\delta t^4) \tag{A6}$$

for an operator $\widetilde{R}_3$ built from nested commutators of A and B.

That local statement alone is not yet a global approximation theorem. The point of Corollary 1 is that one can promote it to a repeated-step estimate without introducing an exponential dependence on the number of steps. Let $W = e^{-i\delta t(A+B)}$ and $\widetilde{W} = V_{\mathrm{LC}}(\delta t)$. Then

$$U(t) - U_N(t) = W^N - \widetilde{W}^N = \sum_{m=0}^{N-1} W^{N-1-m}(W - \widetilde{W})\widetilde{W}^m. \tag{A7}$$

Because both W and $\widetilde{W}$ are unitary, taking the operator norm removes every prefactor except the one-step defect,

$$\|U(t) - U_N(t)\|_2 \le N\|W - \widetilde{W}\|_2. \tag{A8}$$

This is the reason the global error scales as $N\delta t^3 = t^3/N^2$ rather than as a less useful quantity involving geometric growth in N. For the present repository this distinction matters operationally: the method is intended as a low-depth repeated step, so the difference between a stable telescoping argument and a crude worst-case iteration would directly affect how the numerical trends should be read.

Appendix B: Locality of the Nested Commutators

The closure correction is attractive here not only because it cancels the leading BCH term, but also because it stays in a small local Pauli sector. For TFIM, Proposition 2 shows that $i[A, B]$ is a sum of nearest-neighbor YZ and ZY strings. Commuting one more time with the original blocks still preserves short support. Representative identities are

$$\begin{aligned} \left[Z_j Z_{j+1}, Y_j Z_{j+1}\right] &= 2iX_j, \\ \left[X_j, Y_j Z_{j+1}\right] &= 2iZ_j Z_{j+1}, \end{aligned} \tag{B1}$$

and analogous formulas hold for the $Z_j Y_{j+1}$ terms. Thus the third-order remainder entering Corollary 1 does not generate arbitrary long Pauli strings immediately; it remains concentrated in one-, two-, and three-site sectors localized around the original bond support. This helps explain why a one-generator correction can already have visible impact in short static evolutions.

The same qualitative statement persists for XXZ. Here A is again a sum of commuting ZZ bonds, while B is a sum of overlapping $XX + YY$ bond terms. The commutator $[A, B]$ therefore only receives contributions from overlapping bonds, and every nested commutator that enters the third-order BCH remainder is still assembled from operators supported on one bond or on two adjacent bonds. The closure direction is therefore not merely algebraically available; it is still physically local in the sense relevant to low-depth circuit design. That locality would be lost much more quickly in generic dense matrix splittings, where the first commutator can already spread support across the full system.

This locality perspective is one reason the repository benchmark is scientifically defensible despite its small size. The paper is not claiming that all low-depth Lie-closure corrections are useful. It is showing that in models where the closure stays local and interpretable, the first extra generator is often the right one to test before moving to a much richer variational family. In that sense the benchmark isolates a structural question that would be harder to see in larger, less interpretable numerical studies.

Appendix C: Protocol Details and Reproducibility

The entire benchmark lives in a $2^4 \times 2^4$ Hilbert space, so every matrix appearing in the code can be inspected exactly. This matters because the fitted baselines are then compared against exact unitary targets, not against noisy samples or reduced observables. The static TFIM and XXZ models are both encoded as two-block Hamiltonians with explicit term lists, which allows the restricted Cartan baseline to project onto the commuting A-block span and allows LC-QAOA to form the correction from the actual matrix commutator. Nothing in the reported tables is inferred indirectly.

TABLE VI. Protocol summary for the released benchmark scripts.

Benchmark	Parameters	Resolutions
Static TFIM	$n = 4$, selected (J, h, T) cases from released CSVs	$N \in \{2, 4, 6\}$
Static XXZ	$n = 4$, selected (J_{xy}, J_z, T) cases from released CSVs	matched gate proxy
Driven TFIM	$n = 4$, $J = h = 1$, $a \in \{0.5, 1.0\}$, $\omega = 1$, $T \in \{1.0, 2.0\}$	$N \in \{2, 4\}$

The point of Table VI is not to inflate the artifact into a large-scale benchmark. It is to make the limits of the evidence auditable. Every conclusion in the main text can be traced to a small grid of couplings, times, and resolutions that a reader could recompute directly. The gate proxy is similarly transparent. It counts exponentials appearing in the constructed approximation, but it does not include the optimizer cost of fitted baselines or a hardware-native compilation model. That is why the paper consistently phrases comparisons as "matched gate proxy" rather than "matched runtime" or "matched fault-tolerant cost."

The selected-case figures were kept in the main text because they show the shape of the method ordering at a glance, but the aggregate tables matter more scientifically. They show that LC-QAOA is best understood as a targeted structural correction, not a globally dominant heuristic. In the static bank it frequently beats the commuting surrogate, sometimes beats first-order Trotter, and almost never threatens the best optimized or symmetric baselines. In the driven bank it loses the static advantage almost entirely. Those are the right facts to preserve if this repository is later used as a seed for a larger study.

Appendix D: Conference-Literature Taxonomy

The recent conference literature relevant to this paper falls into four themes. The value of stating those themes in prose rather than in a compact comparison table is that the relationship is conceptual, not benchmark-to-benchmark. These papers are not direct baselines for the present repository. They help locate the design principle behind the repository in a much larger conversation about how much mathematically justified structure should be inserted before one turns to a larger model class.

a. Operator learning. Across recent ICML, NeurIPS, and AAAI proceedings, operator-learning papers repeatedly show that architectural choices tied to discretization, geometry, low-rank structure, and conservation laws can produce better approximations than a generic universal approximator used without those constraints [15–23, 25–38]. The present manuscript makes the analogous point for low-depth Hamiltonian simulation. The extra expressive ingredient is not a hidden layer, attention head, or pretraining recipe; it is the first block-commutator direction. Once that framing is explicit, the numerical results become easier to interpret. The correction helps when the first omitted commutator dominates the local defect, and it stops helping when higher-order structure or time dependence takes over.

b. Scientific machine learning. Another relevant thread is the current scientific-ML emphasis on mechanistic priors, physics-informed objectives, and certified surrogates. Recent conference work studies mechanistic networks, parameterized PINNs, Lie-symmetry-aware Gaussian processes, error certification, policy iteration for constrained differential systems, multimodal scientific datasets, and several forms of regularization for physically grounded dynamics [40–46, 48–57]. This paper is much smaller in scope, but it shares the same methodological instinct. When one only has room for a tiny model family, the most useful additional degree of freedom is usually the one demanded by the underlying mathematics rather than the one that merely enlarges the search space.

c. Symmetry and equivariance. The strongest conceptual parallel comes from symmetry-aware modeling. A large 2024–2025 conference literature asks how much performance can be recovered by building equivariance, canonicalization, or symmetry discovery directly into the representation rather than learning it implicitly from data [60–65, 68, 79–83, 85, 90, 91, 95, 96, 102–104, 108–111]. LC-QAOA fits naturally into that worldview. It does not learn a symmetry, but it hard-codes the first nontrivial generator created by the noncommuting split. That is exactly the kind of minimal, interpretable structural move that this literature has made increasingly credible.

d. Quantum-aware ML and hybrid modeling. The quantum-ML literature provides a final point of comparison. Recent conference papers explore quantum data classification, quantum-aware kernels, variational-quantum training heuristics, contextual transport, sampling, optimization, quantum positional encodings, and hybrid chemical reasoning systems [122–129, 131–134]. Those works are again not direct simulation baselines, but they reinforce a useful norm for this manuscript: the right question is not whether one can add more parameters, but whether the next parameter or generator has an interpretable physical role. In the repository studied here, the answer is yes for the first Lie-closure direction and increasingly no once the problem leaves the short-time static regime.

This narrative taxonomy changes the standard by which the paper should be read. A reader focused entirely on asymptotic Hamiltonian-simulation complexity will correctly judge the contribution to be modest. A reader focused on low-depth, structure-aware modeling should judge it differently. The paper identifies a single extra generator, proves why it is the first one worth adding, shows where it helps, and states clearly where it fails. That is a bounded contribution, but it is a real one,

and it is well aligned with current conference trends on structured learning.

One should still be careful not to overread the analogy. Most of the conference papers discussed here optimize expressive models from data, whereas the present manuscript studies exact matrix simulation with no learning loop in the core method. The commonality is methodological: each line of work asks whether a small amount of mathematically motivated structure can outperform a larger but less principled search space. LC-QAOA is the Hamiltonian-simulation instance of that question, and that is why the recent-conference discussion is not just decorative citation expansion. It changes the way the manuscript should be judged without pretending that these conference papers are direct numerical baselines for the repository.